

Lecture 33: Introduction to Bayesian Inference

[This portion is actually meant to be the last part of Lecture 32]

Consider the testing problem $H_0: \sigma^2 = \sigma_0^2$ ag. $H_A: \sigma^2 \neq \sigma_0^2$.
Recall, we have the SLR model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, where $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$

An unbiased estimator of σ^2 is obtained as $\hat{\sigma}^2$, ~~not that~~ where

$$\hat{\sigma}^2 = \frac{SS_{\text{residuals}}}{n-2} \quad \left(\text{Note, } SS_{\text{res}} = \sum \hat{\epsilon}_i^2 = \sum (y_i - \hat{y}_i)^2, \text{ with } \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1 = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} \right)$$

(So, check, $E \sum (y_i - \hat{y}_i)^2 = \sigma^2$, when you're given $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$)

$\Rightarrow y_i$ indep. $N(\beta_0 + \beta_1 x_i, \sigma^2)$
What are $\hat{y}_i$'s distributed as?

The test statistic for the above test is

$$\chi^2 = \frac{(n-2)\hat{\sigma}^2}{\sigma_0^2} \sim \chi_{n-2}^2 \text{ under } H_0 \text{ with critical regions}$$

$$\omega: \text{obsd } \chi^2 > \chi_{n-2, \alpha}^2 \\ < \chi_{n-2, 1-\alpha}^2$$

$$\& \text{obsd } \chi^2 \leq \chi_{n-2, 1-\alpha/2}^2 \text{ or } > \chi_{n-2, \alpha/2}^2$$

For, a $100(1-\alpha)\%$ confidence interval of σ^2 , use

$$P \left[\chi_{n-2, 1-\alpha}^2 \leq \frac{(n-2)\hat{\sigma}^2}{\sigma^2} \leq \chi_{n-2, \alpha}^2 \right] = 1-\alpha$$

$$\Rightarrow 100\% \text{ C.I. is given by } \left[\frac{(n-2)\hat{\sigma}^2}{\chi_{n-2, \alpha}^2}, \frac{(n-2)\hat{\sigma}^2}{\chi_{n-2, 1-\alpha}^2} \right]$$

Lecture 33 starts here.

Recall the Theorem of Total Probability
prob. space, $(\Omega, \mathcal{F}, P)$, $A_1, A_2, \dots$ are mutually exclusive and exhaustive events, i.e.,
 $\forall A_i \in \mathcal{F}, A_i \cap A_j = \emptyset, \bigcup A_i = \Omega$
for any other event $B \in \mathcal{F}$, $P(B) = \sum_{i=1}^{\infty} P(A_i \cap B) = \sum_{i=1}^{\infty} P(A_i) P(B|A_i)$

$P(A_i)$ are prior probs. & $P(B|A_i)$ conditional probs.

Hence, we recall the Bayes' Theorem, as
 given the prior probabilities $P(A_i)$ and $P(B)$,
 (B is an event which happens when at least one of the events A_i 's happens)

we calculate the posterior probabilities, viz.,

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

in terms of the prior probs (usually a finite number of such probs) and the conditional probs

Ex The new method of liquid biopsy test, called Galleri test is being tested for application with following results:

Test result \ True Situation	Benign Tumor	Malignant Tumor
+ve	False +ve	✓
-ve	✓	False -ve

The test detects malignancy with 80% accuracy and benign cases with 90% accuracy.

A person showing 1% chance of malignancy (by traditional method) ends up with a + test result (new method).
 Group of physicians/surgeons take a call of 75% chance of malignancy (putting more faith on the new method).

However, relying on data alone, we see that

$$P(M|+) = \frac{P(+|M) \cdot P(M)}{P(+|M) \cdot P(M) + P(+|B) \cdot P(B)} = \frac{0.8 \times 0.01}{(0.8 \times 0.01) + (0.1 \times 0.99)} \approx 0.075 < 0.75$$

M: malignant
 M^c : not malignant / B: benign
 +: new method +ve result
 -: new method -ve result

Key Difference between classical/frequentist/non-Bayesian and Bayesian approach is while the former group considers parameters as unknown constants, the latter considers them as random variables.

We next move to a random variable with a probability model having one or more ~~const~~ unknown parameters to come closer to a parametric estimation problem.

Ex

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A traffic controller likes to estimate the mean rate of cars passing through a particular crossing for a given time interval.

Frequentist's case: A classic example of Poisson (λ) distn
($X = \#$ cars passing thro' --, $X \sim P(\lambda)$), collect data (frequency of cars — frequentist method) over no. of days to give UMVUE or MLE of λ .

Bayesian approach Prior to collection of data, assume some prior info' on λ , say $\lambda=3$ w.p. 0.7 ($\lambda=3$ more likely) and $\lambda=5$ w.p. 0.3 ($\lambda=5$ less likely), a 2-pt. distn. But instead of using this distn alone to guess an estimate (e.g., an unbiased estimate) of λ , we see whether this prior info' can be improved by using data (actual no. of cars passing --)

If one day, in a randomly selected time interval, observed $x=7$, what are the ~~prob.~~ probs. that $\lambda=3$ and $\lambda=5$.

Using the same prob. model for x (now, $X|\lambda$), i.e., $P(\lambda)$,

we calculate the posterior prob.

$$P(\lambda=3|x=7) = \frac{P(x=7|\lambda=3)P(\lambda=3)}{P(x=7|\lambda=3)P(\lambda=3) + P(x=7|\lambda=5)P(\lambda=5)}$$
$$= \frac{e^{-3} \frac{3^7}{7!} (0.7)}{e^{-3} \frac{3^7}{7!} (0.7) + e^{-5} \frac{5^7}{7!} (0.3)} = 0.328,$$

which goes down from the prior prob. 0.7 because of the data obs'd ($x=7$).

||'ly $P(\lambda=5|x=7) = 1 - 0.328 = 0.672$ increases from the prior prob 0.3 because of the data.